

Telecommunications Case Study

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Exploratory Data Analysis

We conduct an exploratory data analysis by loading the churn dataset and examining its structure, dimensions, and basic statistics. The dataset contains 1,000 rows and 14 columns, with a mix of categorical and numerical data. Summary statistics reveal ranges and central tendencies in the numerical variables, providing an initial view of customer behavior and churn patterns. By running `is.na()`, we also found out that there are no missing data.

```
dataset <- read.csv("churn_dataset.csv", header = TRUE, sep = ",")
dim(dataset)

## [1] 1000  14

str(dataset)

## 'data.frame':  1000 obs. of  14 variables:
## $ Sex          : chr  "Female" "Male" "Female" "Female" ...
## $ Marital_Status : chr  "Married" "Married" "Married" "Married" ...
## $ Term         : int   16 70 36 72 40 15 1 36 5 57 ...
## $ Phone_service : chr  "Yes" "Yes" "Yes" "Yes" ...
## $ International_plan: chr  "Yes" "No" "No" "No" ...
## $ Voice_mail_plan : chr  "Yes" "Yes" "Yes" "No" ...
## $ Multiple_line  : chr  "No" "No" "No" "Yes" ...
## $ Internet_service : chr  "Cable" "Cable" "Cable" "Cable" ...
## $ Technical_support : chr  "Yes" "Yes" "Yes" "Yes" ...
## $ Streaming_Videos : chr  "No" "Yes" "Yes" "Yes" ...
## $ Agreement_period : chr  "Monthly contract" "One year contract"
## "Monthly contract" "One year contract" ...
## $ Monthly_Charges : num   98 75.2 73.3 112.6 95 ...
## $ Total_Charges   : num  1410 5023 2379 7882 3647 ...
## $ Churn           : chr  "Yes" "No" "No" "No" ...

summary(dataset)

##      Sex          Marital_Status      Term      Phone_service
## Length:1000      Length:1000      Min.   : 0.0      Length:1000
## Class :character Class :character 1st Qu.: 8.0      Class :character
## Mode  :character Mode  :character Median :30.0      Mode  :character
##                               Mean  :32.8
##                               3rd Qu.:57.0
##                               Max.   :72.0
## International_plan Voice_mail_plan Multiple_line      Internet_service
## Length:1000      Length:1000      Length:1000      Length:1000
```

```
## Class :character Class :character Class :character Class :character
## Mode :character Mode :character Mode :character Mode :character
##
##
##
## Technical_support Streaming_Videos Agreement_period Monthly_Charges
## Length:1000 Length:1000 Length:1000 Min. : 18.95
## Class :character Class :character Class :character 1st Qu.: 40.16
## Mode :character Mode :character Mode :character Median : 74.72
## Mean : 66.64
## 3rd Qu.: 90.88
## Max. :116.25
##
## Total_Charges Churn
## Min. : 5.0 Length:1000
## 1st Qu.: 334.8 Class :character
## Median :1442.3 Mode :character
## Mean :2351.6
## 3rd Qu.:4016.8
## Max. :8476.5
```

`head(dataset)`

```
## Sex Marital_Status Term Phone_service International_plan
Voice_mail_plan
## 1 Female Married 16 Yes Yes
Yes
## 2 Male Married 70 Yes No
Yes
## 3 Female Married 36 Yes No
Yes
## 4 Female Married 72 Yes No
No
## 5 Female Married 40 Yes Yes
No
## 6 Female Single 15 Yes Yes
Yes
## Multiple_line Internet_service Technical_support Streaming_Videos
## 1 No Cable Yes No
## 2 No Cable Yes Yes
## 3 No Cable Yes Yes
## 4 Yes Cable Yes Yes
## 5 Yes Cable No Yes
## 6 No No Internet No internet No internet
## Agreement_period Monthly_Charges Total_Charges Churn
## 1 Monthly contract 98.05 1410.25 Yes
## 2 One year contract 75.25 5023.00 No
## 3 Monthly contract 73.35 2379.10 No
## 4 One year contract 112.60 7882.25 No
## 5 Monthly contract 95.05 3646.80 No
## 6 Monthly contract 19.85 255.35 No
```

```
tail(dataset)
```

```
##           Sex Marital_Status Term Phone_service International_plan
## 995      Male         Single  59           No              No
## 996   Female       Married  56           Yes              No
## 997   Female       Married  56           Yes              No
## 998      Male       Married  57           Yes              No
## 999      Male       Married  51           Yes              No
## 1000     Male         Single  69           Yes             yes
##           Voice_mail_plan Multiple_line Internet_service Technical_support
## 995                No      No phone          DSL             Yes
## 996                Yes        Yes      Fiber optic           No
## 997                Yes        Yes          DSL             No
## 998                Yes        No      Fiber optic           No
## 999                Yes        No          DSL             Yes
## 1000               Yes        No      No Internet      No internet
##           Streaming_Videos Agreement_period Monthly_Charges Total_Charges
Churn
## 995                No One year contract          44.45      2145.00
No
## 996                Yes One year contract          84.50      4054.20
No
## 997                Yes Two year contract          83.50      3958.25
No
## 998                No One year contract          73.95      4326.80
No
## 999                No One year contract          60.90      3582.40
No
## 1000      No internet Two year contract          19.15      1363.45
No
```

```
names(dataset)
```

```
## [1] "Sex"           "Marital_Status" "Term"
## [4] "Phone_service" "International_plan" "Voice_mail_plan"
## [7] "Multiple_line" "Internet_service" "Technical_support"
## [10] "Streaming_Videos" "Agreement_period" "Monthly_Charges"
## [13] "Total_Charges" "Churn"
```

```
sum(is.na(dataset))
```

```
## [1] 0
```

Univariate Analysis

Categorical Variables

For our univariate analysis, we first focus on categorical variables to understand their individual distributions. We start by isolating the categorical columns, excluding numerical ones like `Term`, `Monthly\_Charges`, and `Total\_Charges`. Using `lapply()` with `table()`, we summarize each categorical variable, gaining insights into customer

characteristics such as gender, marital status, and service plans. For instance, there are slightly more male customers (514) than female (486), and most customers are married (735). Service-related attributes like `Phone\_service` and `Internet\_service` also show patterns: 911 customers have phone service, while internet services vary, with fiber optic being the most popular.

We then visualize these distributions with bar graphs for each categorical variable using `ggplot2`. By iterating over each column and applying `geom\_bar()`, we create a consistent and clear visual overview, which helps us quickly interpret the data presented.

```
categorical_data <- dataset[, -c(3, 12, 13)]
```

```
lapply(categorical_data, table)
```

```
## $Sex
##
## Female    Male
##    486    514
##
## $Marital_Status
##
## Married   Single
##    735    265
##
## $Phone_service
##
## No Yes
##  89 911
##
## $International_plan
##
## No yes Yes
## 429 262 309
##
## $Voice_mail_plan
##
## No Yes
## 442 558
##
## $Multiple_line
##
##           No No phone           Yes
##           467           89         444
##
## $Internet_service
##
```

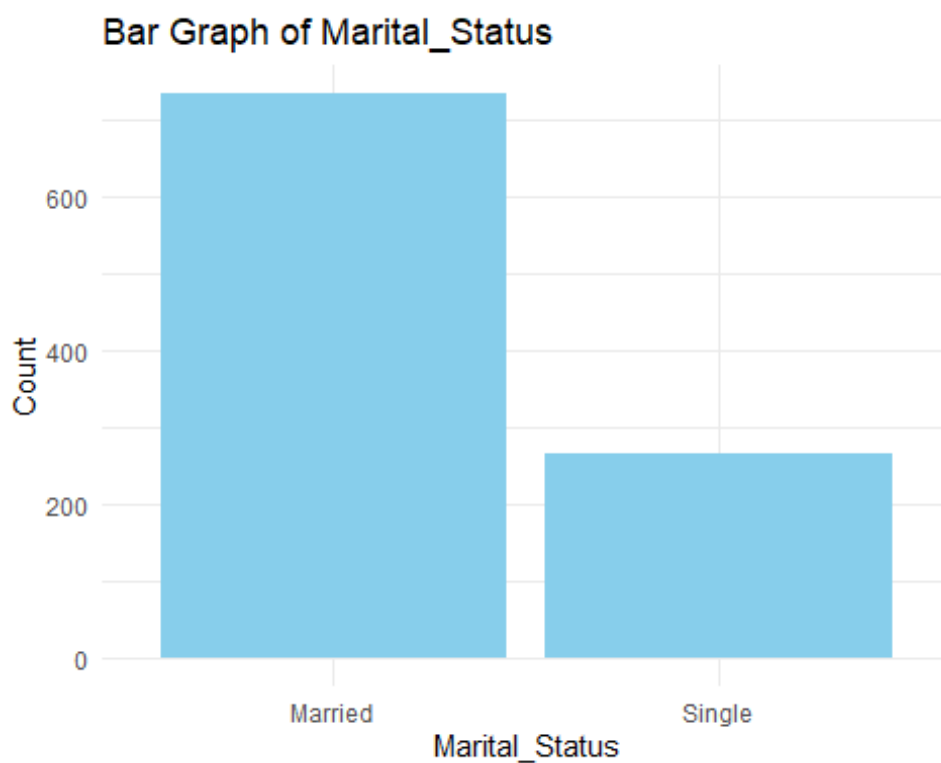
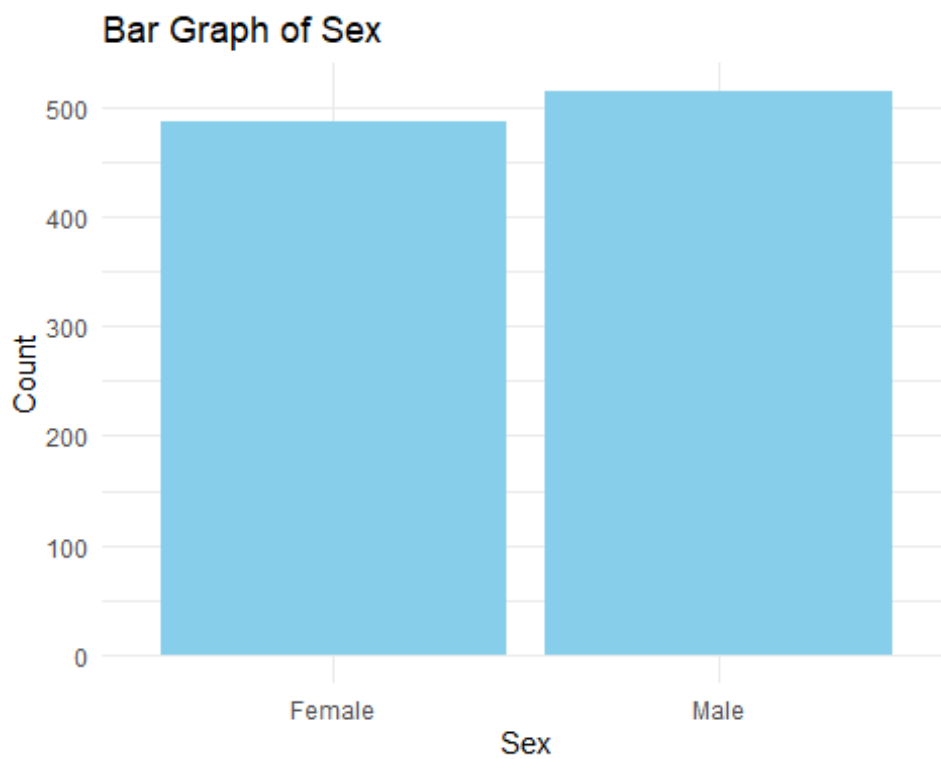
```
##      Cable      DSL Fiber optic No Internet
##      171      280      341      208
##
## $Technical_support
##
##      No No internet      Yes
##      476      208      316
##
## $Streaming_Videos
##
##      No No internet      Yes
##      382      208      410
##
## $Agreement_period
##
## Monthly contract One year contract Two year contract
##      539      208      253
##
## $Churn
##
## No Yes
## 741 259
```

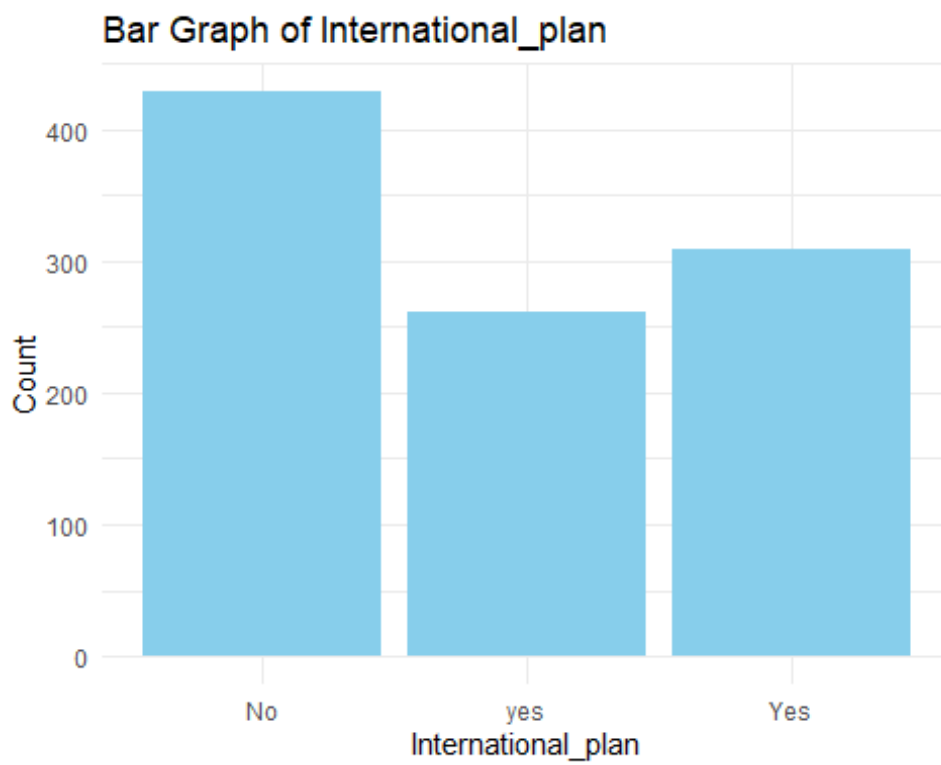
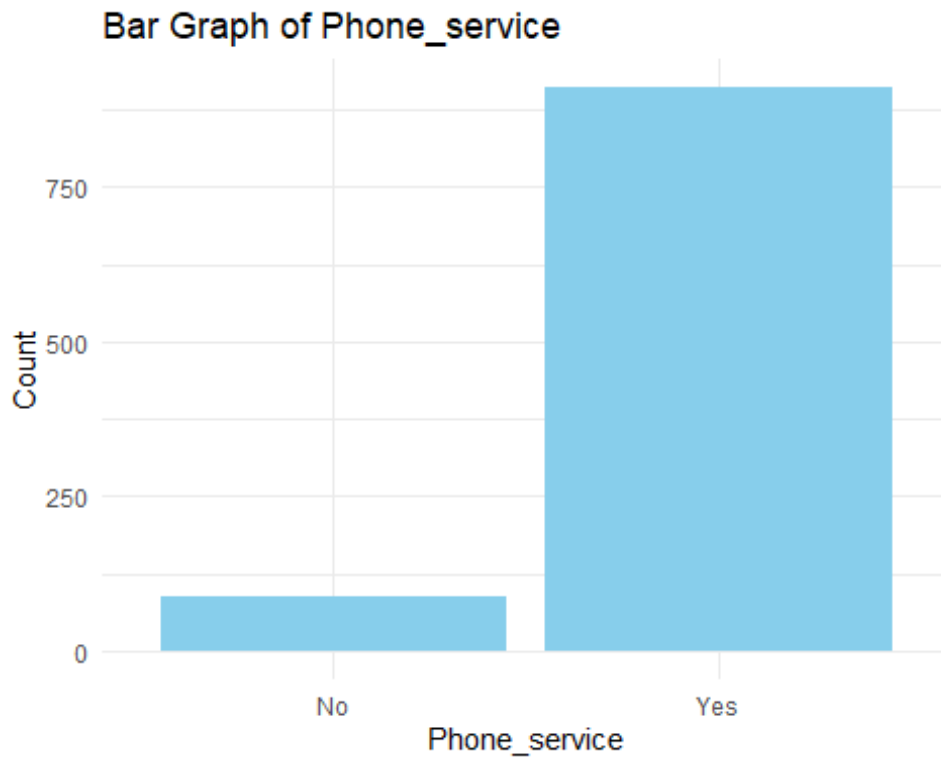
```
library(ggplot2)
```

```
univariate_plots <- lapply(names(categorical_data), function(col_name) {
  ggplot(categorical_data, aes_string(x = col_name)) +
    geom_bar(fill = "skyblue") +
    labs(title = paste("Bar Graph of", col_name), x = col_name, y = "Count")
  +
    theme_minimal()
})
```

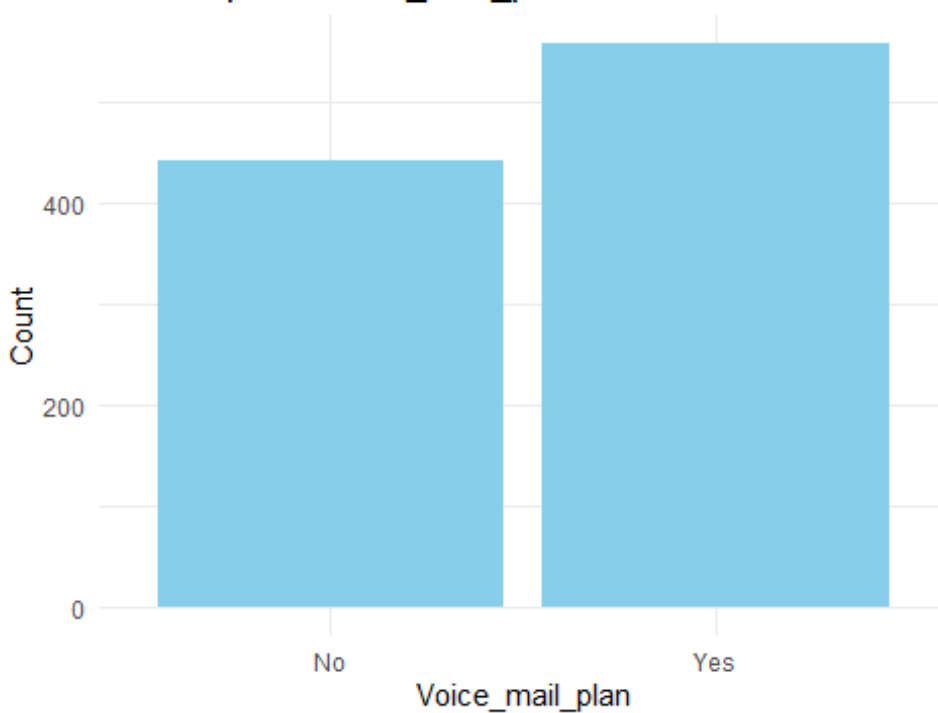
```
## Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
## i Please use tidy evaluation idioms with `aes()`.
## i See also `vignette("ggplot2-in-packages")` for more information.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
for (plot in univariate_plots) {
  print(plot)
}
```

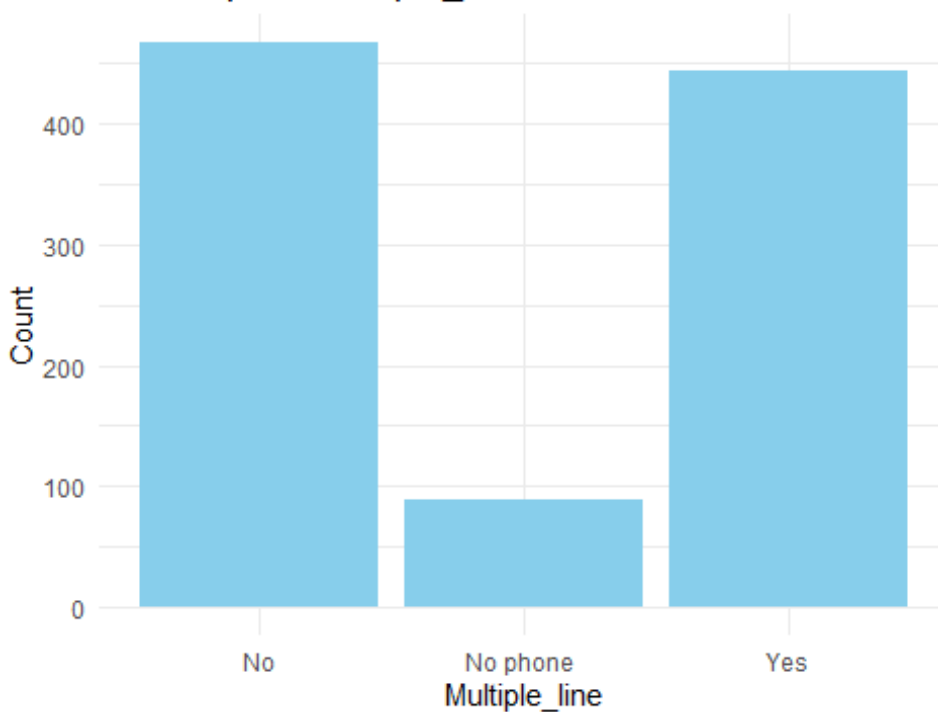


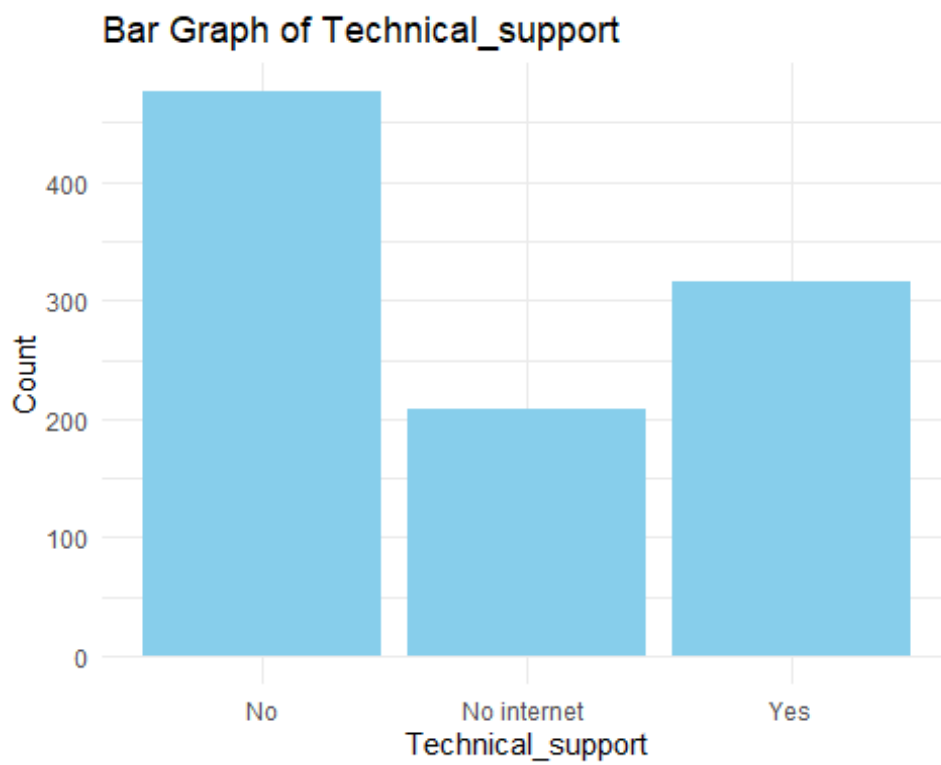
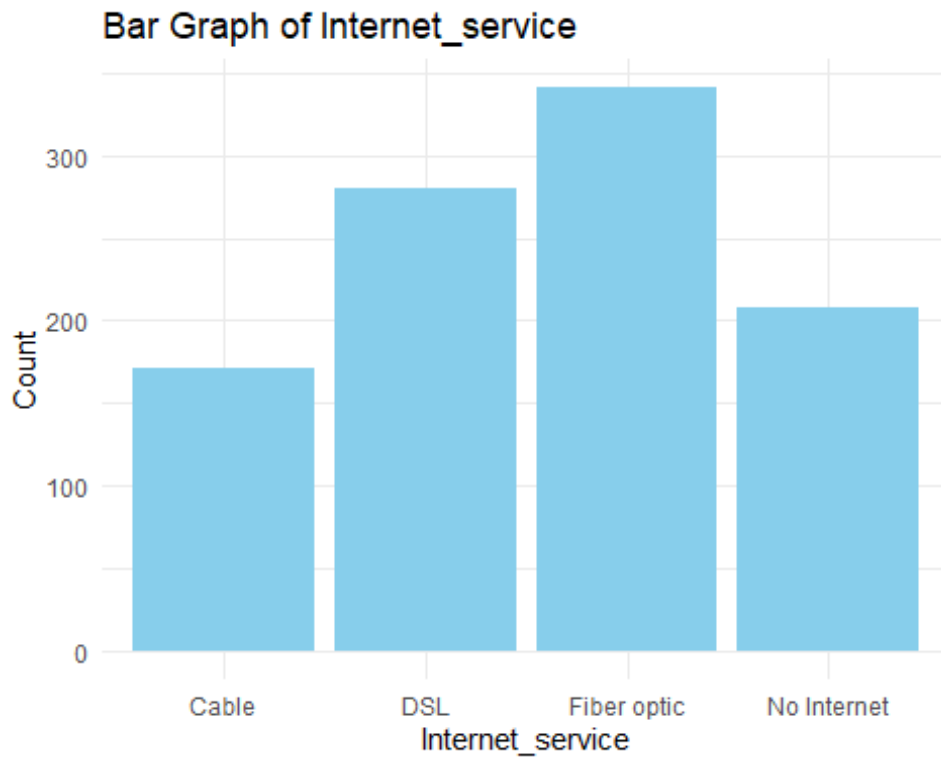


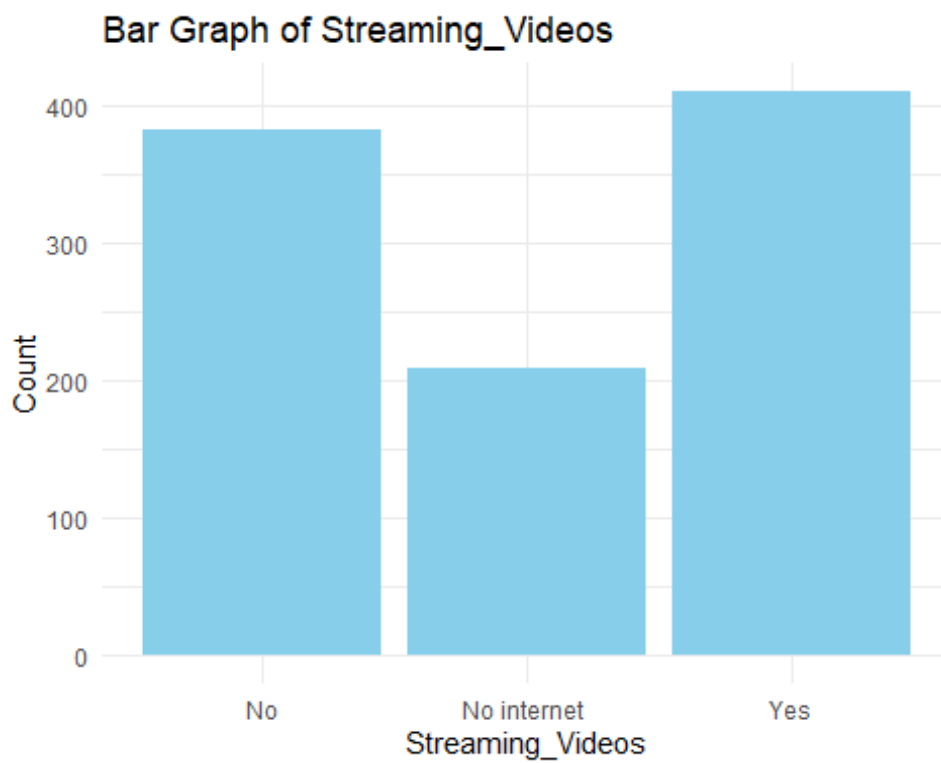
Bar Graph of Voice_mail_plan

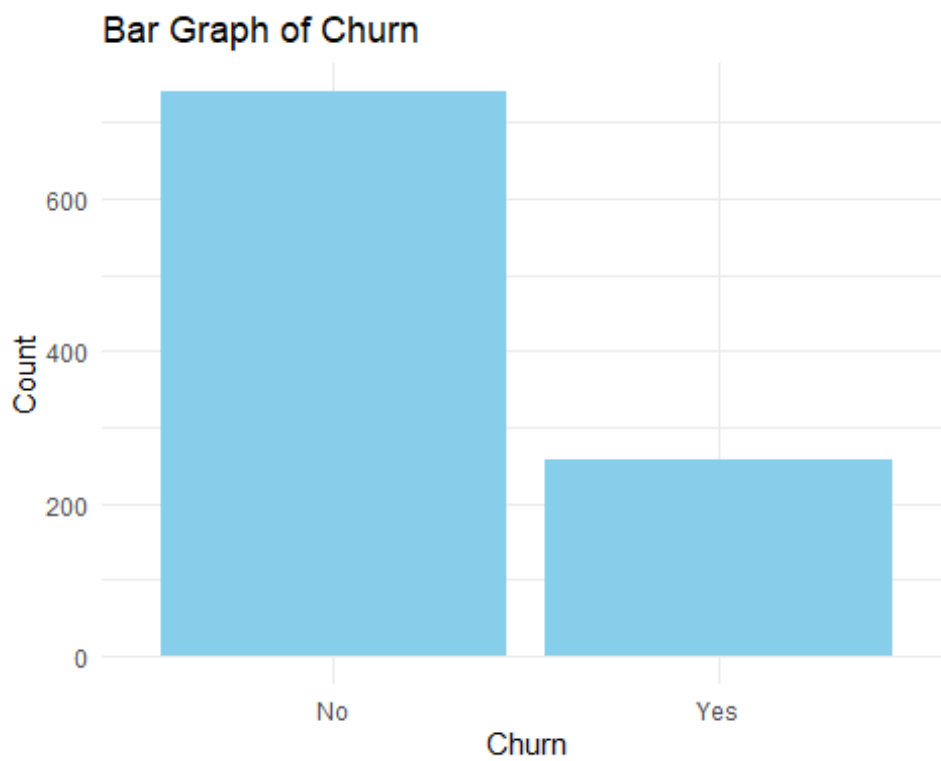
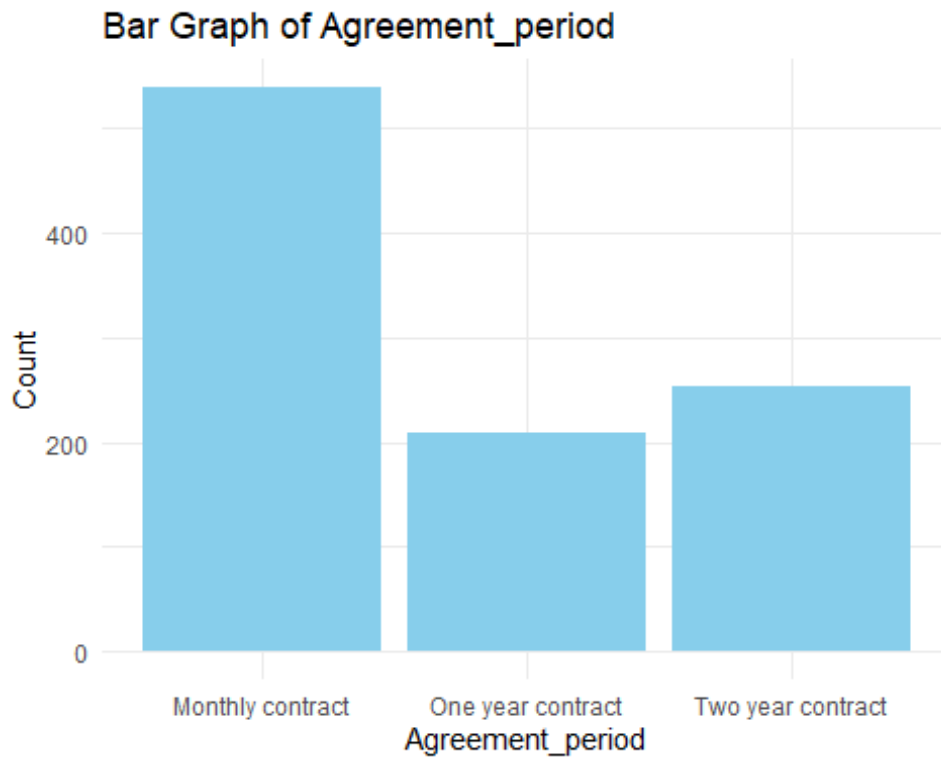


Bar Graph of Multiple_line









Numerical Variables

Proceeding to our univariate analysis of numerical variables, we examine `Term`, `Monthly\_Charges`, and `Total\_Charges` to assess their distributions and basic statistics. We use `summary()` to gather key metrics, including the minimum, first quartile, median, mean, third quartile, and maximum values, which provide a foundational understanding of the range and central tendencies within each variable. For example, `Term` ranges from 0 to 72 months, with a median of 30 months, while `Monthly\_Charges` has a median of \$74.72, reflecting variability in customer billing.

Additionally, we apply the Shapiro-Wilk test to assess normality, revealing that each variable deviates significantly from a normal distribution (p-value < 0.05). Calculating the IQR and standard deviation gives further insights into the spread of each variable, helping identify potential outliers. Using IQR-based thresholds, we define lower and upper bounds for each variable, aiding in the identification of values that may fall outside typical ranges.

To visualize these distributions, we create histograms for each numerical variable using `ggplot2`, adding a dashed line for the mean (in orange) and IQR-based bounds (in dark blue). This provides a quick overview of the data's spread and helps in identifying patterns, such as skewness or the presence of outliers.

```
numerical_data <- dataset[, c(3,12,13)]

for (col_name in names(numerical_data)) {
  column_data <- numerical_data[[col_name]]

  summary_stats <- summary(column_data)
  cat("\nSummary of", col_name, ":\n")
  print(summary_stats)

  shapiro_test <- shapiro.test(column_data)
  cat("\nShapiro-Wilk Test for", col_name, ":\n")
  print(shapiro_test)

  iqr_value <- IQR(column_data)
  cat("\nIQR of", col_name, ":", iqr_value, "\n")

  sd_value <- sd(column_data)
  cat("\nStandard Deviation of", col_name, ":", sd_value, "\n")

  Q1 <- summary_stats["1st Qu."]
  Q3 <- summary_stats["3rd Qu."]
  cat("\nQ1 (1st Quartile) of", col_name, ":", Q1, "\n")
  cat("\nQ3 (3rd Quartile) of", col_name, ":", Q3, "\n")

  lower_bound <- Q1 - 1.5 * iqr_value
```

```

upper_bound <- Q3 + 1.5 * iqr_value
cat("\nLower Bound for", col_name, ":", lower_bound, "\n")
cat("Upper Bound for", col_name, ":", upper_bound, "\n")
}

```

```

##
## Summary of Term :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   0.0      8.0     30.0     32.8    57.0     72.0
##
## Shapiro-Wilk Test for Term :
##
##  Shapiro-Wilk normality test
##
## data:  column_data
## W = 0.89544, p-value < 2.2e-16
##
##
## IQR of Term : 49
##
## Standard Deviation of Term : 25.13565
##
## Q1 (1st Quartile) of Term : 8
## Q3 (3rd Quartile) of Term : 57
##
## Lower Bound for Term : -65.5
## Upper Bound for Term : 130.5
##
## Summary of Monthly_Charges :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  18.95  40.16   74.72   66.64  90.88  116.25
##
## Shapiro-Wilk Test for Monthly_Charges :
##
##  Shapiro-Wilk normality test
##
## data:  column_data
## W = 0.91288, p-value < 2.2e-16
##
##
## IQR of Monthly_Charges : 50.7125
##
## Standard Deviation of Monthly_Charges : 30.29861
##
## Q1 (1st Quartile) of Monthly_Charges : 40.1625
## Q3 (3rd Quartile) of Monthly_Charges : 90.875
##
## Lower Bound for Monthly_Charges : -35.90625
## Upper Bound for Monthly_Charges : 166.9437
##

```

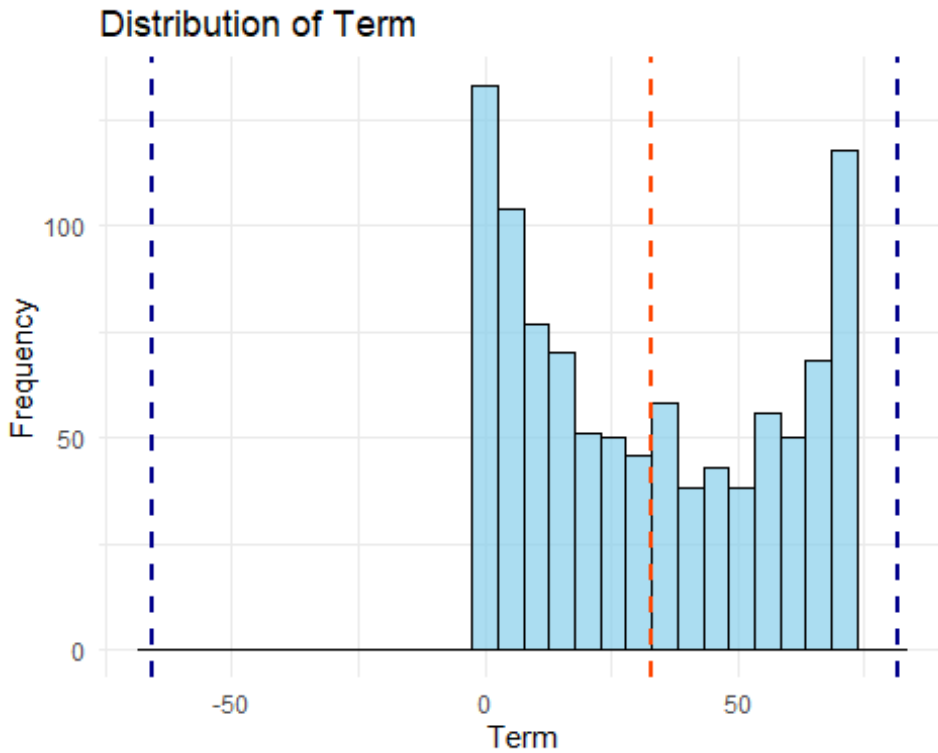
```

## Summary of Total_Charges :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     5.0   334.8  1442.3  2351.6  4016.8  8476.5
##
## Shapiro-Wilk Test for Total_Charges :
##
##   Shapiro-Wilk normality test
##
## data:  column_data
## W = 0.8663, p-value < 2.2e-16
##
##
## IQR of Total_Charges : 3681.95
##
## Standard Deviation of Total_Charges : 2318.049
##
## Q1 (1st Quartile) of Total_Charges : 334.8
## Q3 (3rd Quartile) of Total_Charges : 4016.75
##
## Lower Bound for Total_Charges : -5188.125
## Upper Bound for Total_Charges : 9539.675

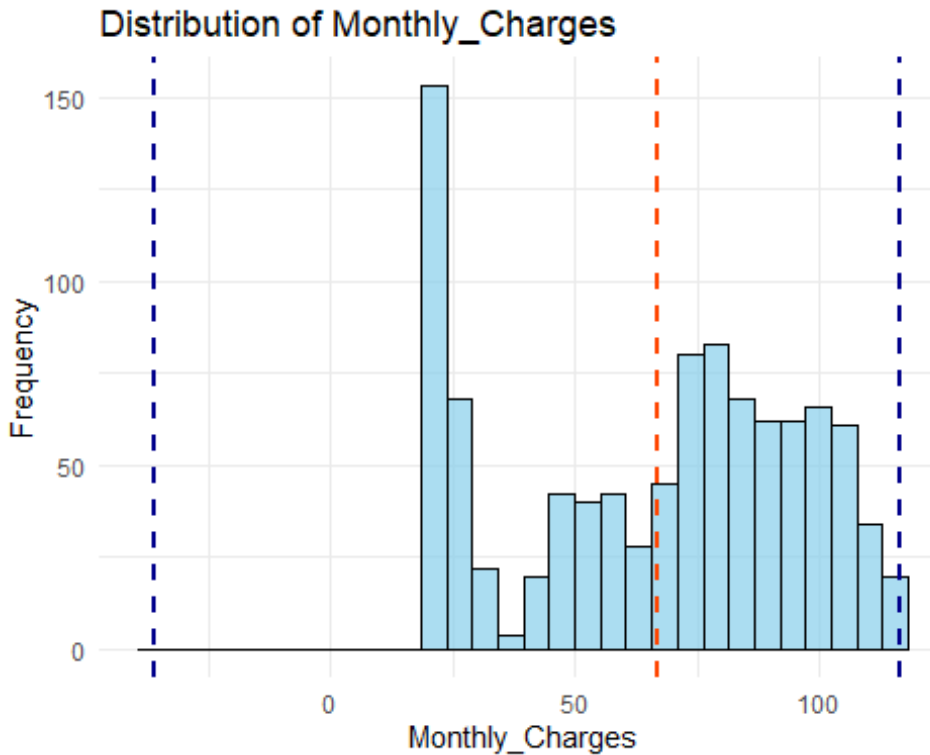
ggplot(dataset, aes(x = Term)) +
  geom_histogram(bins = 30, fill = "skyblue", color = "black", alpha = 0.7) +
  labs(title = "Distribution of Term", x = "Term", y =
"Frequency") +
  theme_minimal() +
  geom_vline(aes(xintercept = mean(Term)), color = "orangered", linetype =
"dashed", size = 1) +
  geom_vline(
    aes(xintercept = quantile(Term, 0.25) - 1.5 * IQR(Term)),
    color = "darkblue", linetype = "dashed", size = 1
  ) +
  geom_vline(
    aes(xintercept = quantile(Term, 0.25) + 1.5 * IQR(Term)),
    color = "darkblue", linetype = "dashed", size = 1
  )
)

## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

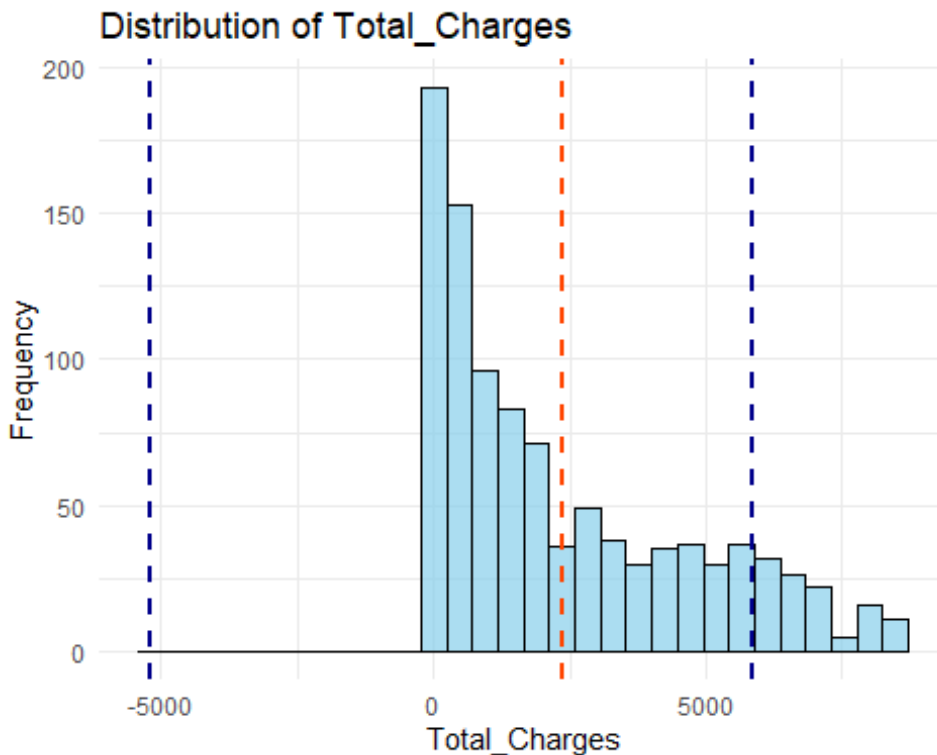
```



```
ggplot(dataset, aes(x = Monthly_Charges)) +
  geom_histogram(bins = 30, fill = "skyblue", color = "black", alpha = 0.7) +
  labs(title = "Distribution of Monthly_Charges", x = "Monthly_Charges", y =
"Frequency") +
  theme_minimal() +
  geom_vline(aes(xintercept = mean(Monthly_Charges)), color = "orangered",
linetype = "dashed", size = 1) +
  geom_vline(
    aes(xintercept = quantile(Monthly_Charges, 0.25) - 1.5 *
IQR(Monthly_Charges)),
    color = "darkblue", linetype = "dashed", size = 1
  ) +
  geom_vline(
    aes(xintercept = quantile(Monthly_Charges, 0.25) + 1.5 *
IQR(Monthly_Charges)),
    color = "darkblue", linetype = "dashed", size = 1
  )
)
```



```
ggplot(dataset, aes(x = Total_Charges)) +
  geom_histogram(bins = 30, fill = "skyblue", color = "black", alpha = 0.7) +
  labs(title = "Distribution of Total_Charges", x = "Total_Charges", y =
"Frequency") +
  theme_minimal() +
  geom_vline(aes(xintercept = mean(Total_Charges)), color = "orangered",
linetype = "dashed", size = 1) +
  geom_vline(
    aes(xintercept = quantile(Total_Charges, 0.25) - 1.5 *
IQR(Total_Charges)),
    color = "darkblue", linetype = "dashed", size = 1
  ) +
  geom_vline(
    aes(xintercept = quantile(Total_Charges, 0.25) + 1.5 *
IQR(Total_Charges)),
    color = "darkblue", linetype = "dashed", size = 1
  )
)
```



Bivariate Analysis

Categorical x Categorical

The bivariate analysis explores the relationships between `Churn` and the categorical variables `Sex`, `Marital\_Status`, and `Agreement\_period`. Examining `Sex` and `Churn` through a contingency table reveals similar churn rates between male and female customers. A Chi-squared test confirms this lack of significant association, with a p-value of 0.1249, indicating that customer churn does not significantly differ by gender. However, a different pattern emerges when looking at `Marital\_Status`. Married customers are less likely to churn compared to single customers, and the Chi-squared test provides a p-value of 4.19e-06, showing a statistically significant association. This result suggests that marital status could play a role in customer loyalty, with single customers having a higher tendency to churn.

The strongest relationship appears when analyzing `Agreement\_period` and `Churn`. Customers with monthly contracts exhibit the highest churn rates, while those with longer-term contracts (one or two years) tend to remain. The Chi-squared test yields a highly significant p-value ($< 2.2e-16$), indicating a strong correlation between the contract type and churn behavior. This analysis suggests that marital status and agreement period are both relevant factors for churn, which can be useful for developing more focused customer retention strategies.

```
contingency_sex_churn <- table(dataset$Sex, dataset$Churn)
cat("\nContingency Table: Sex vs Churn\n")
```

```
##
## Contingency Table: Sex vs Churn

print(contingency_sex_churn)

##
##           No Yes
## Female 349 137
## Male   392 122

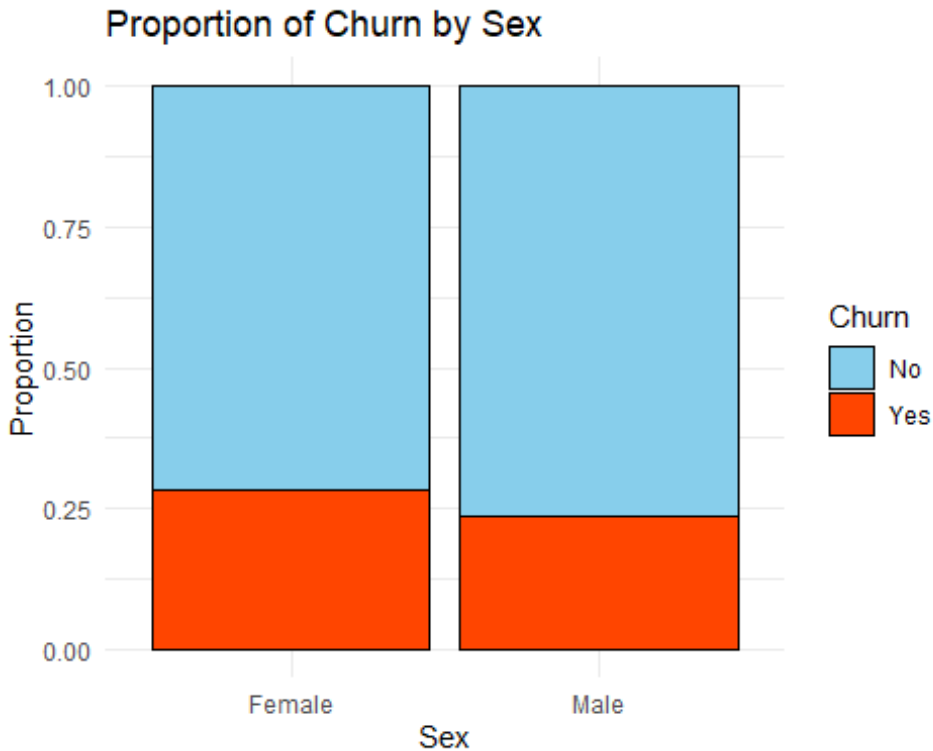
chi_sex_churn <- chisq.test(contingency_sex_churn)
cat("\nChi-squared Test: Sex vs Churn\n")

##
## Chi-squared Test: Sex vs Churn

print(chi_sex_churn)

##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data:  contingency_sex_churn
## X-squared = 2.3552, df = 1, p-value = 0.1249

ggplot(dataset, aes(x = Sex, fill = Churn)) +
  geom_bar(position = "fill", color = "black") +
  labs(title = "Proportion of Churn by Sex", x = "Sex", y = "Proportion") +
  scale_fill_manual(values = c("Yes" = "orangered", "No" = "skyblue")) +
  theme_minimal()
```



```
contingency_marital_churn <- table(dataset$Marital_Status, dataset$Churn)
cat("\nContingency Table: Marital Status vs Churn\n")

##
## Contingency Table: Marital Status vs Churn
print(contingency_marital_churn)

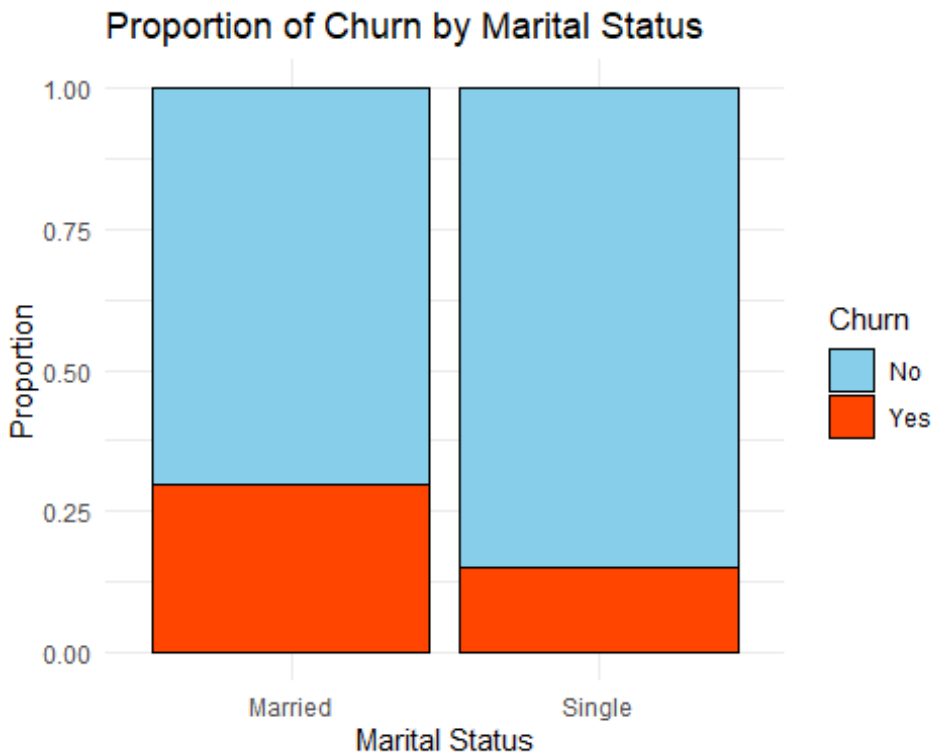
##
##           No Yes
## Married  516 219
## Single   225  40

chi_marital_churn <- chisq.test(contingency_marital_churn)
cat("\nChi-squared Test: Marital Status vs Churn\n")

##
## Chi-squared Test: Marital Status vs Churn
print(chi_marital_churn)

##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data:  contingency_marital_churn
## X-squared = 21.176, df = 1, p-value = 4.19e-06
```

```
ggplot(dataset, aes(x = Marital_Status, fill = Churn)) +
  geom_bar(position = "fill", color = "black") +
  labs(title = "Proportion of Churn by Marital Status", x = "Marital Status",
y = "Proportion") +
  scale_fill_manual(values = c("Yes" = "orangered", "No" = "skyblue")) +
  theme_minimal()
```



```
contingency_agreement_churn <- table(dataset$Agreement_period, dataset$Churn)
cat("\nContingency Table: Agreement Period vs Churn\n")
```

```
##
## Contingency Table: Agreement Period vs Churn
```

```
print(contingency_agreement_churn)
```

```
##
##           No Yes
## Monthly contract 310 229
## One year contract 190 18
## Two year contract 241 12
```

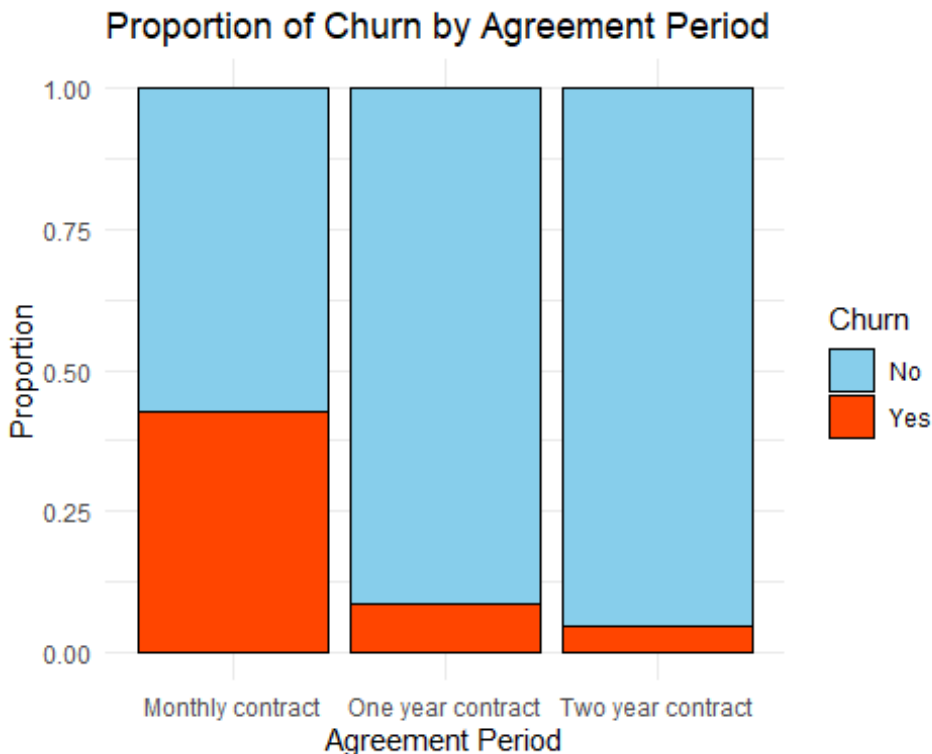
```
chi_agreement_churn <- chisq.test(contingency_agreement_churn)
cat("\nChi-squared Test: Agreement Period vs Churn\n")
```

```
##
## Chi-squared Test: Agreement Period vs Churn
```

```
print(chi_agreement_churn)
```

```
##
## Pearson's Chi-squared test
##
## data: contingency_agreement_churn
## X-squared = 168.5, df = 2, p-value < 2.2e-16

ggplot(dataset, aes(x = Agreement_period, fill = Churn)) +
  geom_bar(position = "fill", color = "black") +
  labs(title = "Proportion of Churn by Agreement Period", x = "Agreement
Period", y = "Proportion") +
  scale_fill_manual(values = c("Yes" = "orangered", "No" = "skyblue")) +
  theme_minimal()
```



Numerical x Categorical

The analysis explores the relationship between `Churn` and numerical variables. For `Term`, the average duration is significantly higher for non-churned customers (37.1) compared to churned customers (20.6), with a highly significant ANOVA result (p-value < 2e-16). Boxplots illustrate this clear difference between the two groups, showing that non-churned customers tend to have longer terms. Additionally, for `Agreement\_period` and `Monthly\_Charges`, the analysis reveals no significant differences between contract types (p-value = 0.252), with all three contract periods showing similar average charges.

In terms of `Total\_Charges`, non-churned customers have a higher average (2558) than churned customers (1762), supported by a significant ANOVA result (p-value = 1.76e-06). Boxplots further confirm this trend, with non-churned customers consistently

exhibiting higher total charges. These findings suggest that churn is associated with lower terms and total charges, while contract types have minimal impact on monthly charges.

```
library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

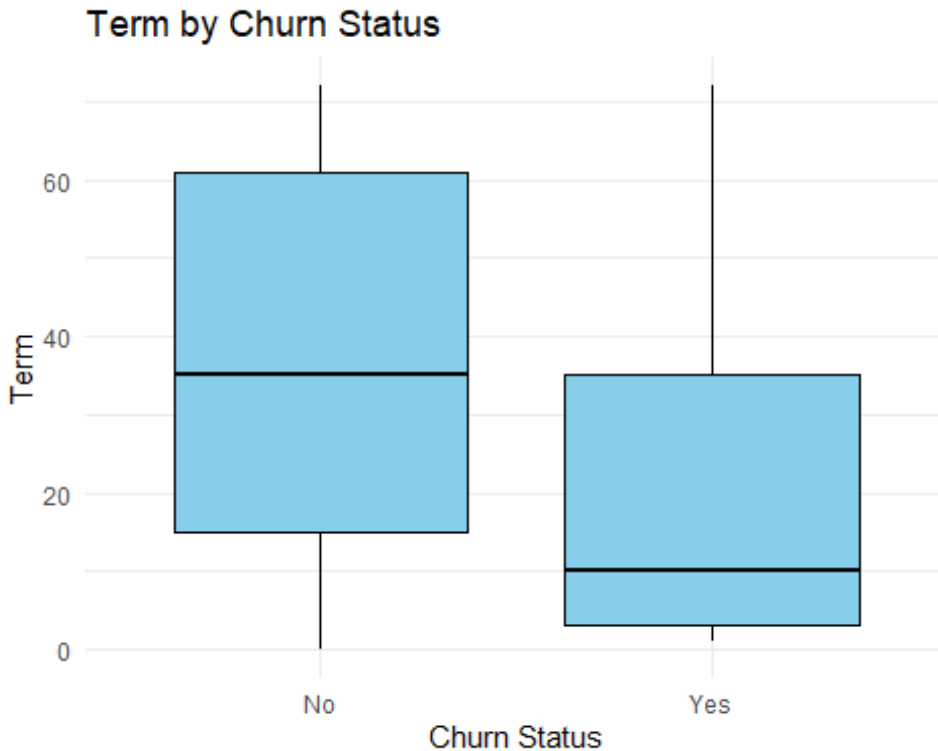
dataset %>%
  group_by(Churn) %>%
  summarise(
    Mean_Term = mean(Term, na.rm = TRUE),
    SD_Term = sd(Term, na.rm = TRUE),
    Min_Term = min(Term, na.rm = TRUE),
    Max_Term = max(Term, na.rm = TRUE)
  )

## # A tibble: 2 × 5
##   Churn Mean_Term SD_Term Min_Term Max_Term
##   <chr>    <dbl>   <dbl>   <int>   <int>
## 1 No      37.1    24.8       0      72
## 2 Yes     20.6    21.8       1      72

anova_term_churn <- aov(Term ~ Churn, data = dataset)
summary(anova_term_churn)

##              Df Sum Sq Mean Sq F value Pr(>F)
## Churn          1  51816   51816   89.26 <2e-16 ***
## Residuals     998 579353     581
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

ggplot(dataset, aes(x = Churn, y = Term)) +
  geom_boxplot(fill = "skyblue", color = "black") +
  labs(title = "Term by Churn Status", x = "Churn Status", y = "Term") +
  theme_minimal()
```



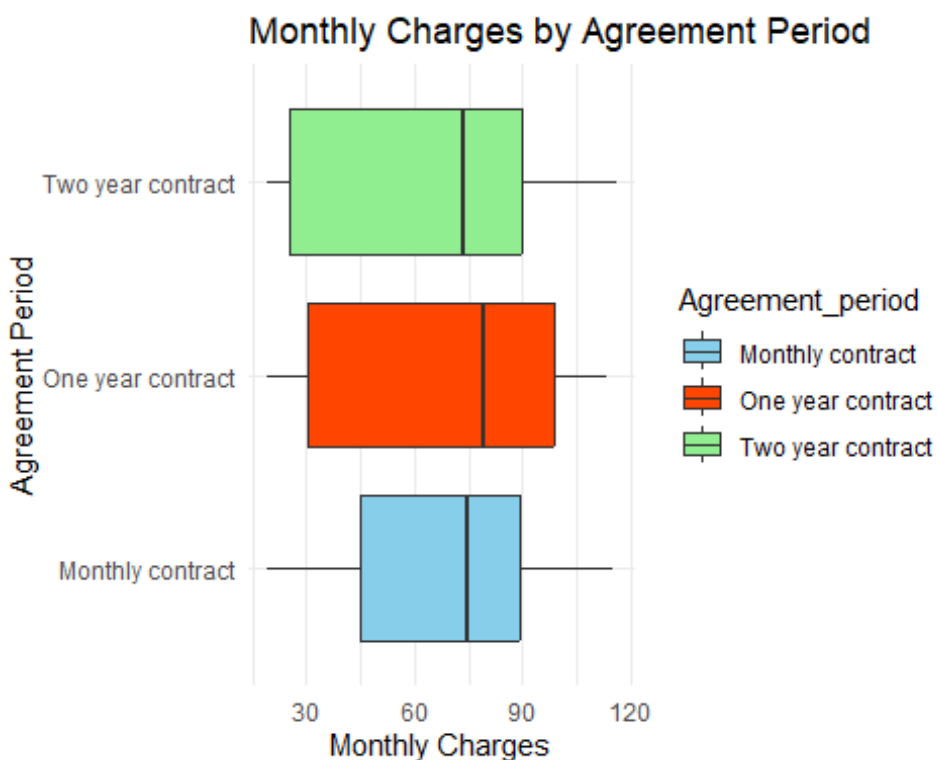
```
dataset %>%
  group_by(Agreement_period) %>%
  summarise(
    Mean_Monthly_Charges = mean(Monthly_Charges, na.rm = TRUE),
    SD_Monthly_Charges = sd(Monthly_Charges, na.rm = TRUE),
    Min_Monthly_Charges = min(Monthly_Charges, na.rm = TRUE),
    Max_Monthly_Charges = max(Monthly_Charges, na.rm = TRUE)
  )

## # A tibble: 3 × 5
##   Agreement_period Mean_Monthly_Charges SD_Monthly_Charges
##   <chr>              <dbl>              <dbl>
## 1 Monthly contract      66.8              27.8
## 2 One year contract     69.0              32.6
## 3 Two year contract     64.3              33.3
## # i 1 more variable: Max_Monthly_Charges <dbl>

anova_agreement_period_charges <- aov(Monthly_Charges ~ Agreement_period,
data = dataset)
summary(anova_agreement_period_charges)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## Agreement_period  2   2536   1267.8    1.382   0.252
## Residuals       997 914552    917.3
```

```
ggplot(dataset, aes(x = Agreement_period, y = Monthly_Charges, fill =
Agreement_period)) +
  geom_boxplot() +
  labs(title = "Monthly Charges by Agreement Period", x = "Agreement Period",
y = "Monthly Charges") +
  scale_fill_manual(values = c("Two year contract" = "lightgreen", "Monthly
contract" = "skyblue", "One year contract" = "orangered")) +
  theme_minimal() +
  coord_flip()
```



```
dataset %>%
  group_by(Churn) %>%
  summarise(
    Mean_Total_Charges = mean(Total_Charges, na.rm = TRUE),
    SD_Total_Charges = sd(Total_Charges, na.rm = TRUE),
    Min_Total_Charges = min(Total_Charges, na.rm = TRUE),
    Max_Total_Charges = max(Total_Charges, na.rm = TRUE)
  )

## # A tibble: 2 × 5
##   Churn Mean_Total_Charges SD_Total_Charges Min_Total_Charges
##   <chr>           <dbl>           <dbl>           <dbl>
```

```

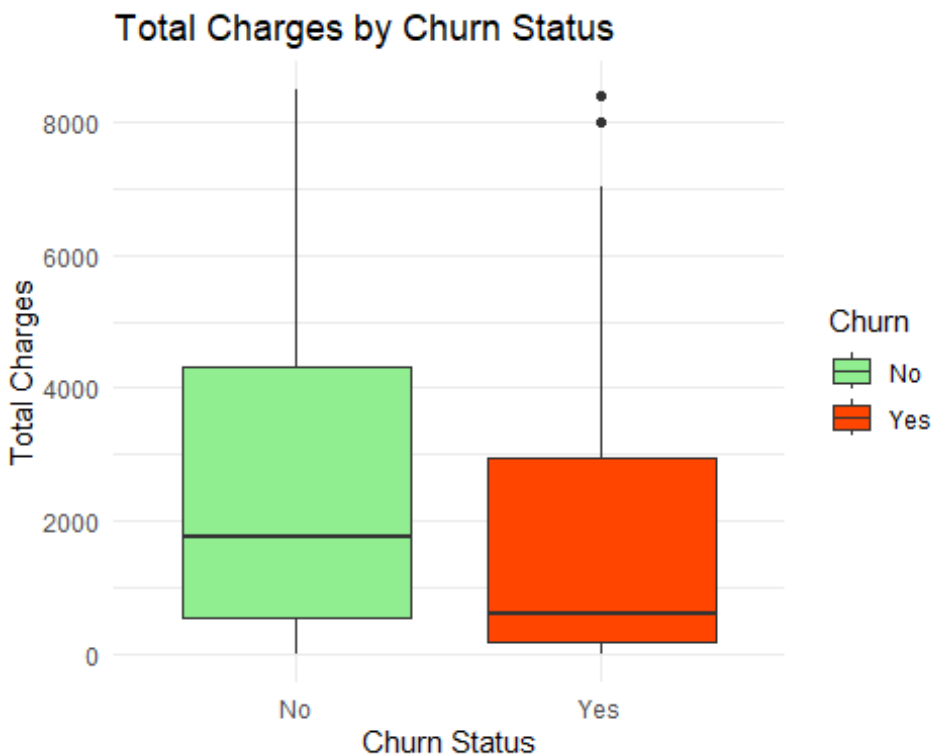
<dbl>
## 1 No                2558.          2358.           5
8476.
## 2 Yes               1762.          2096.          19
8399.

anova_total_charges_churn <- aov(Total_Charges ~ Churn, data = dataset)
summary(anova_total_charges_churn)

##              Df      Sum Sq   Mean Sq F value    Pr(>F)
## Churn          1 1.215e+08 121538275    23.12 1.76e-06 ***
## Residuals     998 5.246e+09   5256956
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

ggplot(dataset, aes(x = Churn, y = Total_Charges, fill = Churn)) +
  geom_boxplot() +
  labs(title = "Total Charges by Churn Status", x = "Churn Status", y =
"Total Charges") +
  scale_fill_manual(values = c("Yes" = "orangered", "No" = "lightgreen")) +
  theme_minimal()

```



Numerical x Numerical

The analysis examines the relationships between numerical variables through correlation and linear regression. The correlation between `Term` and `Monthly\_Charges` is 0.259, suggesting a weak positive relationship. The linear regression model indicates that for each additional month in `Term`, `Monthly\_Charges` increase by approximately 0.31. This relationship is statistically significant (p-value < 2e-16), with an R-squared value of 0.067, meaning the model explains just 6.7% of the variation in `Monthly\_Charges`. The scatter plot with the regression line visually supports this weak positive correlation.

The correlation between `Term` and `Total\_Charges` is much stronger at 0.83, indicating a strong positive relationship. The linear regression reveals that each additional month in `Term` corresponds to an increase of 76.58 in `Total\_Charges`, with a highly significant p-value (< 2e-16) and an R-squared value of 0.6895. This suggests that the model accounts for 69% of the variation in `Total\_Charges`. Additionally, the correlation between `Total\_Charges` and `Monthly\_Charges` is 0.63, indicating a moderate positive relationship. The regression model shows that for every increase in `Total\_Charges`, `Monthly\_Charges` rise by 0.008, with an R-squared value of 0.392, explaining about 39% of the variability in `Monthly\_Charges`.

```
cor_term_monthly_charges <- cor(dataset$Term, dataset$Monthly_Charges, use =
"complete.obs")
cat("\nCorrelation between Term and Monthly Charges: ",
cor_term_monthly_charges, "\n")

##
## Correlation between Term and Monthly Charges: 0.2585645

lm_term_monthly_charges <- lm(Monthly_Charges ~ Term, data = dataset)
cat("\nLinear Regression (Term vs Monthly Charges): \n")

##
## Linear Regression (Term vs Monthly Charges):

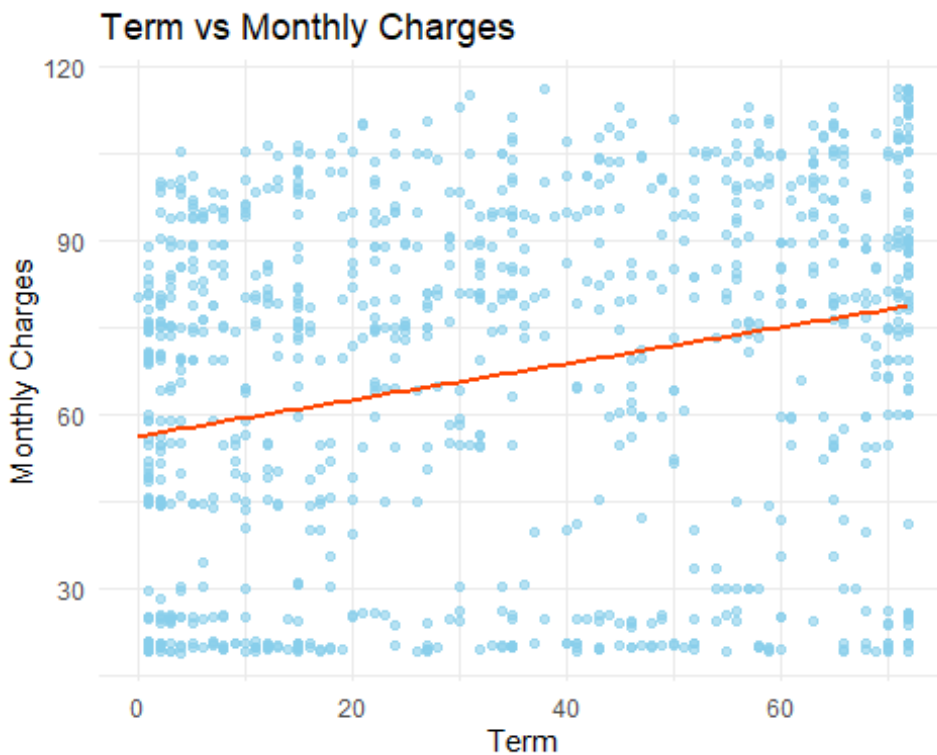
print(summary(lm_term_monthly_charges))

##
## Call:
## lm(formula = Monthly_Charges ~ Term, data = dataset)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -59.455 -28.508   8.218  24.660  49.023
```

```
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 56.41486    1.52301  37.042  <2e-16 ***
## Term        0.31167     0.03686   8.456  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 29.28 on 998 degrees of freedom
## Multiple R-squared:  0.06686,    Adjusted R-squared:  0.06592
## F-statistic: 71.5 on 1 and 998 DF,  p-value: < 2.2e-16

ggplot(dataset, aes(x = Term, y = Monthly_Charges)) +
  geom_point(color = "skyblue", alpha = 0.6) +
  labs(title = "Term vs Monthly Charges", x = "Term", y = "Monthly Charges")
+
  theme_minimal() +
  geom_smooth(method = "lm", se = FALSE, color = "orangered")

## `geom_smooth()` using formula = 'y ~ x'
```



```
cor_term_total_charges <- cor(dataset$Term, dataset$Total_Charges, use =
"complete.obs")
cat("\nCorrelation between Term and Total Charges: ", cor_term_total_charges,
"\n")
```

```
##
## Correlation between Term and Total Charges: 0.8303744

lm_term_total_charges <- lm(Total_Charges ~ Term, data = dataset)
cat("\nLinear Regression (Term vs Total Charges): \n")

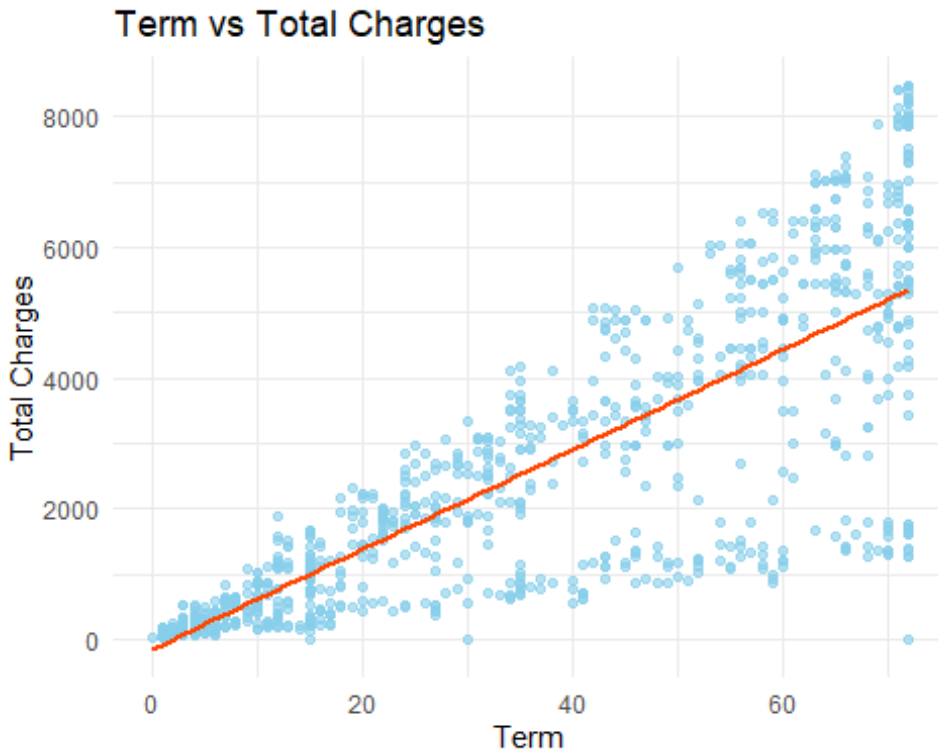
##
## Linear Regression (Term vs Total Charges):

print(summary(lm_term_total_charges))

##
## Call:
## lm(formula = Total_Charges ~ Term, data = dataset)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5348.1  -315.2   125.5   537.1  3128.4
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -160.555      67.211  -2.389   0.0171 *
## Term         76.578       1.627  47.079 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1292 on 998 degrees of freedom
## Multiple R-squared:  0.6895, Adjusted R-squared:  0.6892
## F-statistic: 2216 on 1 and 998 DF, p-value: < 2.2e-16

ggplot(dataset, aes(x = Term, y = Total_Charges)) +
  geom_point(color = "skyblue", alpha = 0.6) +
  labs(title = "Term vs Total Charges", x = "Term", y = "Total Charges") +
  theme_minimal() +
  geom_smooth(method = "lm", se = FALSE, color = "orangered")

## `geom_smooth()` using formula = 'y ~ x'
```



```
cor_total_charges_monthly_charges <- cor(dataset$Total_Charges,
dataset$Monthly_Charges, use = "complete.obs")
cat("\nCorrelation between Total Charges and Monthly Charges: ",
cor_total_charges_monthly_charges, "\n")

##
## Correlation between Total Charges and Monthly Charges: 0.626028

lm_total_charges_monthly_charges <- lm(Monthly_Charges ~ Total_Charges, data
= dataset)
cat("\nLinear Regression (Total Charges vs Monthly Charges): \n")

##
## Linear Regression (Total Charges vs Monthly Charges):

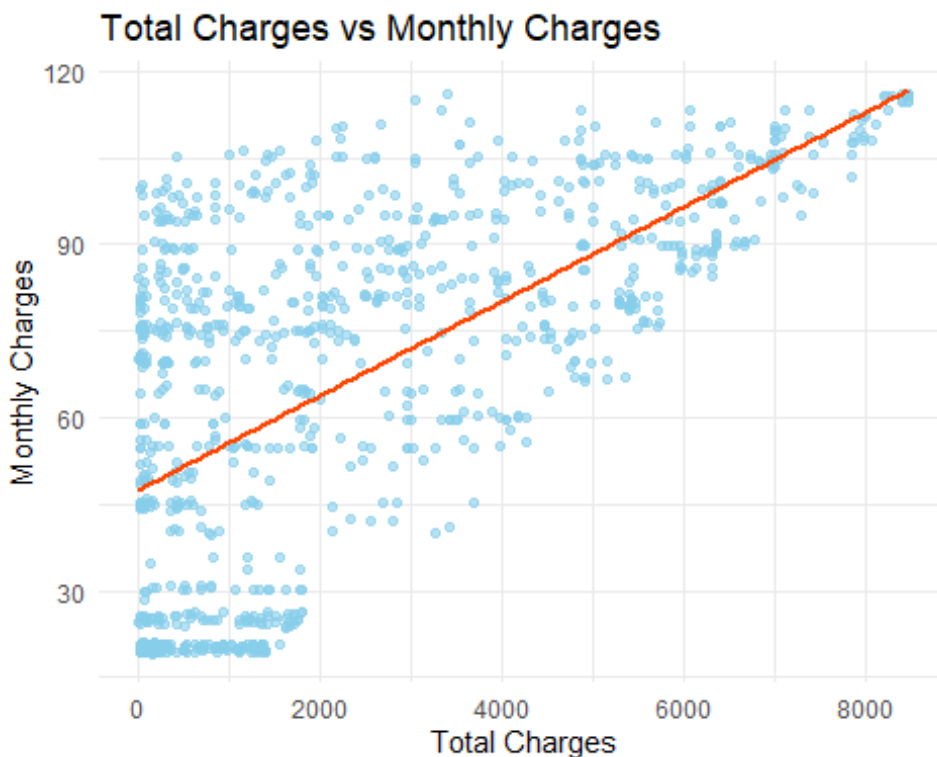
print(summary(lm_total_charges_monthly_charges))

##
## Call:
## lm(formula = Monthly_Charges ~ Total_Charges, data = dataset)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -39.668 -21.970  -1.122  18.558  54.480
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)  4.740e+01  1.065e+00  44.50  <2e-16 ***
## Total_Charges 8.183e-03  3.226e-04  25.36  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 23.64 on 998 degrees of freedom
## Multiple R-squared:  0.3919, Adjusted R-squared:  0.3913
## F-statistic: 643.2 on 1 and 998 DF,  p-value: < 2.2e-16

ggplot(dataset, aes(x = Total_Charges, y = Monthly_Charges)) +
  geom_point(color = "skyblue", alpha = 0.6) +
  labs(title = "Total Charges vs Monthly Charges", x = "Total Charges", y =
"Monthly Charges") +
  theme_minimal() +
  geom_smooth(method = "lm", se = FALSE, color = "orangered")

## `geom_smooth()` using formula = 'y ~ x'
```



Model Building

```
library(rpart)
library(caret)

## Loading required package: lattice

library(e1071)
library(caTools)
library(rpart.plot)
library(RColorBrewer)
library(pROC)

## Type 'citation("pROC")' for a citation.

##
## Attaching package: 'pROC'

## The following objects are masked from 'package:stats':
##
##     cov, smooth, var

library(MLmetrics)

##
## Attaching package: 'MLmetrics'

## The following objects are masked from 'package:caret':
##
##     MAE, RMSE

## The following object is masked from 'package:base':
##
##     Recall

dataset$Churn <- factor(dataset$Churn, levels = c("No", "Yes"))
set.seed(123)
trainIndex <- createDataPartition(dataset$Churn, p = 0.7, list = FALSE)
train_data <- dataset[trainIndex, ]
test_data <- dataset[-trainIndex, ]

dt_model <- rpart(Churn ~ ., data = train_data, method = "class")
summary(dt_model)

## Call:
## rpart(formula = Churn ~ ., data = train_data, method = "class")
##     n= 701
```

```

##
##          CP nsplit rel error    xerror      xstd
## 1 0.05677656      0 1.0000000 1.0000000 0.06378066
## 2 0.01648352      5 0.6208791 0.6648352 0.05497640
## 3 0.01282051      7 0.5879121 0.6813187 0.05550963
## 4 0.01000000     10 0.5494505 0.7472527 0.05752588
##
## Variable importance
##      Total_Charges  Agreement_period      Term
Monthly_Charges
##              18              16              16
13
## Technical_support  Streaming_Videos  Internet_service
Marital_Status
##              11              7              6
4
## International_plan  Multiple_line  Voice_mail_plan
Sex
##              4              2              2
2
##
## Node number 1: 701 observations,    complexity param=0.05677656
## predicted class=No expected loss=0.2596291 P(node) =1
## class counts:  519  182
## probabilities: 0.740 0.260
## left son=2 (319 obs) right son=3 (382 obs)
## Primary splits:
##      Agreement_period splits as RLL,      improve=39.80786, (0
missing)
##      Technical_support splits as RLL,      improve=30.25102, (0
missing)
##      Internet_service splits as RLRL,      improve=18.74216, (0
missing)
##      Monthly_Charges < 30.8 to the left, improve=17.81895, (0
missing)
##      Term < 16.5 to the right, improve=17.31430, (0
missing)
## Surrogate splits:
##      Term < 35.5 to the right, agree=0.815, adj=0.592,
(0 split)
##      Total_Charges < 3656.3 to the right, agree=0.713, adj=0.370,
(0 split)
##      Technical_support splits as RLL,      agree=0.693, adj=0.326,
(0 split)
##      International_plan splits as LRR,      agree=0.622, adj=0.169,
(0 split)
##      Streaming_Videos splits as RLL,      agree=0.609, adj=0.141,
(0 split)
##
## Node number 2: 319 observations

```

```

## predicted class=No expected loss=0.07523511 P(node) =0.4550642
## class counts: 295 24
## probabilities: 0.925 0.075
##
## Node number 3: 382 observations, complexity param=0.05677656
## predicted class=No expected loss=0.4136126 P(node) =0.5449358
## class counts: 224 158
## probabilities: 0.586 0.414
## left son=6 (75 obs) right son=7 (307 obs)
## Primary splits:
## Monthly_Charges < 41.725 to the left, improve=17.584900, (0
missing)
## Internet_service splits as RRRL, improve=12.740050, (0
missing)
## Technical_support splits as RLR, improve=12.740050, (0
missing)
## Streaming_Videos splits as RLR, improve=12.740050, (0
missing)
## Marital_Status splits as RL, improve= 6.864863, (0
missing)
## Surrogate splits:
## Internet_service splits as RRRL, agree=0.953, adj=0.760, (0 split)
## Technical_support splits as RLR, agree=0.953, adj=0.760, (0 split)
## Streaming_Videos splits as RLR, agree=0.953, adj=0.760, (0 split)
## Marital_Status splits as RL, agree=0.893, adj=0.453, (0 split)
## Phone_service splits as LR, agree=0.814, adj=0.053, (0 split)
##
## Node number 6: 75 observations
## predicted class=No expected loss=0.1066667 P(node) =0.10699
## class counts: 67 8
## probabilities: 0.893 0.107
##
## Node number 7: 307 observations, complexity param=0.05677656
## predicted class=No expected loss=0.4885993 P(node) =0.4379458
## class counts: 157 150
## probabilities: 0.511 0.489
## left son=14 (236 obs) right son=15 (71 obs)
## Primary splits:
## Monthly_Charges < 94.55 to the left, improve=9.747123, (0
missing)
## Term < 18 to the right, improve=7.143966, (0
missing)
## Total_Charges < 802.15 to the right, improve=6.802518, (0
missing)
## Technical_support splits as R-L, improve=2.027639, (0
missing)
## Marital_Status splits as LR, improve=1.065590, (0
missing)
## Surrogate splits:
## Total_Charges < 5968.6 to the left, agree=0.805, adj=0.155, (0

```

```

split)
##      Term                < 56.5    to the left,  agree=0.775, adj=0.028, (0
split)
##      Streaming_Videos splits as  L-R,            agree=0.772, adj=0.014, (0
split)
##
## Node number 14: 236 observations,    complexity param=0.05677656
##   predicted class=No   expected loss=0.4194915  P(node) =0.3366619
##   class counts:   137    99
##   probabilities: 0.581 0.419
##   left son=28 (110 obs) right son=29 (126 obs)
##   Primary splits:
##     Total_Charges    < 802.15  to the right, improve=15.224950, (0
missing)
##     Term              < 12.5    to the right, improve=12.196350, (0
missing)
##     Monthly_Charges  < 50.125  to the right, improve= 4.089567, (0
missing)
##     Marital_Status   splits as  LR,            improve= 2.670025, (0
missing)
##     Multiple_line    splits as  RLL,           improve= 1.745400, (0
missing)
##   Surrogate splits:
##     Term              < 10.5    to the right, agree=0.915, adj=0.818,
(0 split)
##     Multiple_line    splits as  RRL,           agree=0.674, adj=0.300,
(0 split)
##     Monthly_Charges  < 75.95   to the right, agree=0.653, adj=0.255,
(0 split)
##     International_plan splits as  LRR,         agree=0.614, adj=0.173,
(0 split)
##     Marital_Status   splits as  LR,            agree=0.589, adj=0.118,
(0 split)
##
## Node number 15: 71 observations,    complexity param=0.01648352
##   predicted class=Yes  expected loss=0.2816901  P(node) =0.1012839
##   class counts:    20    51
##   probabilities: 0.282 0.718
##   left son=30 (21 obs) right son=31 (50 obs)
##   Primary splits:
##     Technical_support splits as  R-L,           improve=2.256204, (0
missing)
##     Monthly_Charges  < 106.675 to the right, improve=2.098442, (0
missing)
##     Term              < 17.5    to the right, improve=1.780267, (0
missing)
##     Total_Charges    < 1776.75 to the right, improve=1.780267, (0
missing)
##     Voice_mail_plan  splits as  LR,            improve=1.222717, (0
missing)

```

```

## Surrogate splits:
##   Monthly_Charges < 101.6   to the right, agree=0.761, adj=0.19, (0
split)
##
## Node number 28: 110 observations
##   predicted class=No   expected loss=0.2272727   P(node) =0.1569187
##   class counts:      85    25
##   probabilities: 0.773 0.227
##
## Node number 29: 126 observations,   complexity param=0.05677656
##   predicted class=Yes   expected loss=0.4126984   P(node) =0.1797432
##   class counts:       52    74
##   probabilities: 0.413 0.587
##   left son=58 (26 obs) right son=59 (100 obs)
##   Primary splits:
##   Total_Charges      < 44.825   to the left,   improve=10.2224400, (0
missing)
##   Technical_support splits as  R-L,           improve= 2.1616780, (0
missing)
##   Term               < 4.5      to the left,   improve= 1.3028580, (0
missing)
##   Monthly_Charges   < 76.275   to the left,   improve= 0.7404904, (0
missing)
##   Multiple_line     splits as  LLR,           improve= 0.6606230, (0
missing)
##
## Node number 30: 21 observations,   complexity param=0.01648352
##   predicted class=Yes   expected loss=0.4761905   P(node) =0.0299572
##   class counts:       10    11
##   probabilities: 0.476 0.524
##   left son=60 (10 obs) right son=61 (11 obs)
##   Primary splits:
##   Voice_mail_plan   splits as  LR,           improve=4.0034630, (0
missing)
##   Internet_service splits as  R-L-,          improve=1.3223440, (0
missing)
##   Monthly_Charges   < 104.325 to the left,   improve=1.1904760, (0
missing)
##   Term              < 23       to the right, improve=0.7619048, (0
missing)
##   Total_Charges     < 2201.55 to the right, improve=0.7619048, (0
missing)
##   Surrogate splits:
##   Internet_service splits as  R-L-,          agree=0.762, adj=0.5, (0
split)
##   Sex               splits as  RL,           agree=0.667, adj=0.3, (0
split)
##   Term              < 17.5     to the right, agree=0.667, adj=0.3, (0
split)
##   Multiple_line     splits as  L-R,          agree=0.667, adj=0.3, (0

```

```

split)
##      Monthly_Charges < 104.325 to the left, agree=0.667, adj=0.3, (0
split)
##
## Node number 31: 50 observations
##   predicted class=Yes expected loss=0.2 P(node) =0.07132668
##   class counts:    10    40
##   probabilities: 0.200 0.800
##
## Node number 58: 26 observations
##   predicted class=No expected loss=0.1923077 P(node) =0.03708987
##   class counts:    21     5
##   probabilities: 0.808 0.192
##
## Node number 59: 100 observations, complexity param=0.01282051
##   predicted class=Yes expected loss=0.31 P(node) =0.1426534
##   class counts:    31    69
##   probabilities: 0.310 0.690
##   left son=118 (79 obs) right son=119 (21 obs)
##   Primary splits:
##       Total_Charges < 90.775 to the right, improve=2.452092, (0
missing)
##       Technical_support splits as R-L, improve=1.972062, (0
missing)
##       Term < 1.5 to the right, improve=1.303810, (0
missing)
##       Multiple_line splits as LLR, improve=1.085393, (0
missing)
##       Monthly_Charges < 90.25 to the right, improve=1.028848, (0
missing)
##   Surrogate splits:
##       Term < 2.5 to the right, agree=0.88, adj=0.429, (0 split)
##
## Node number 60: 10 observations
##   predicted class=No expected loss=0.2 P(node) =0.01426534
##   class counts:     8     2
##   probabilities: 0.800 0.200
##
## Node number 61: 11 observations
##   predicted class=Yes expected loss=0.1818182 P(node) =0.01569187
##   class counts:     2     9
##   probabilities: 0.182 0.818
##
## Node number 118: 79 observations, complexity param=0.01282051
##   predicted class=Yes expected loss=0.3670886 P(node) =0.1126961
##   class counts:    29    50
##   probabilities: 0.367 0.633
##   left son=236 (26 obs) right son=237 (53 obs)
##   Primary splits:
##       Total_Charges < 297.7 to the left, improve=2.2763500, (0

```

```

missing)
##      Multiple_line      splits as  LLR,          improve=1.2941780, (0
missing)
##      Monthly_Charges   < 76.025  to the left,  improve=1.1892530, (0
missing)
##      Technical_support splits as  R-L,          improve=1.0234440, (0
missing)
##      Voice_mail_plan   splits as  LR,          improve=0.9437773, (0
missing)
##      Surrogate splits:
##      Term              < 2.5      to the left,  agree=0.785, adj=0.346,
(0 split)
##      International_plan splits as  RRL,          agree=0.684, adj=0.038,
(0 split)
##
## Node number 119: 21 observations
## predicted class=Yes expected loss=0.0952381 P(node) =0.0299572
## class counts:      2    19
## probabilities: 0.095 0.905
##
## Node number 236: 26 observations,      complexity param=0.01282051
## predicted class=No expected loss=0.4615385 P(node) =0.03708987
## class counts:     14    12
## probabilities: 0.538 0.462
## left son=472 (17 obs) right son=473 (9 obs)
## Primary splits:
##      Sex              splits as  LR,          improve=2.7531420, (0
missing)
##      Total_Charges    < 164.5    to the right, improve=1.9230770, (0
missing)
##      Monthly_Charges < 62.125    to the right, improve=1.8480770, (0
missing)
##      Term             < 2.5      to the right, improve=1.1583710, (0
missing)
##      Voice_mail_plan splits as  LR,          improve=0.8480769, (0
missing)
##      Surrogate splits:
##      Term < 1.5       to the right, agree=0.692, adj=0.111, (0 split)
##
## Node number 237: 53 observations
## predicted class=Yes expected loss=0.2830189 P(node) =0.07560628
## class counts:     15    38
## probabilities: 0.283 0.717
##
## Node number 472: 17 observations
## predicted class=No expected loss=0.2941176 P(node) =0.02425107
## class counts:     12     5
## probabilities: 0.706 0.294
##
## Node number 473: 9 observations

```

```

## predicted class=Yes expected loss=0.222222 P(node) =0.0128388
## class counts:      2      7
## probabilities: 0.222 0.778

predictions <- predict(dt_model, test_data, type = "class")

conf_matrix <- confusionMatrix(predictions, test_data$Churn)
print(conf_matrix)

## Confusion Matrix and Statistics
##
##              Reference
## Prediction   No  Yes
##      No    207  37
##      Yes    15  40
##
##              Accuracy : 0.8261
##              95% CI : (0.7783, 0.8673)
##      No Information Rate : 0.7425
##      P-Value [Acc > NIR] : 0.0003848
##
##              Kappa : 0.4984
##
##  Mcnemar's Test P-Value : 0.0035892
##
##              Sensitivity : 0.9324
##              Specificity : 0.5195
##              Pos Pred Value : 0.8484
##              Neg Pred Value : 0.7273
##              Prevalence : 0.7425
##              Detection Rate : 0.6923
##              Detection Prevalence : 0.8161
##              Balanced Accuracy : 0.7260
##
##              'Positive' Class : No
##

accuracy <- conf_matrix$overall['Accuracy']
precision <- conf_matrix$byClass['Pos Pred Value']
recall <- conf_matrix$byClass['Sensitivity']
f1_score <- 2 * (precision * recall) / (precision + recall)

cat("\nModel Evaluation Metrics:\n")

##
## Model Evaluation Metrics:

cat("Accuracy: ", accuracy, "\n")

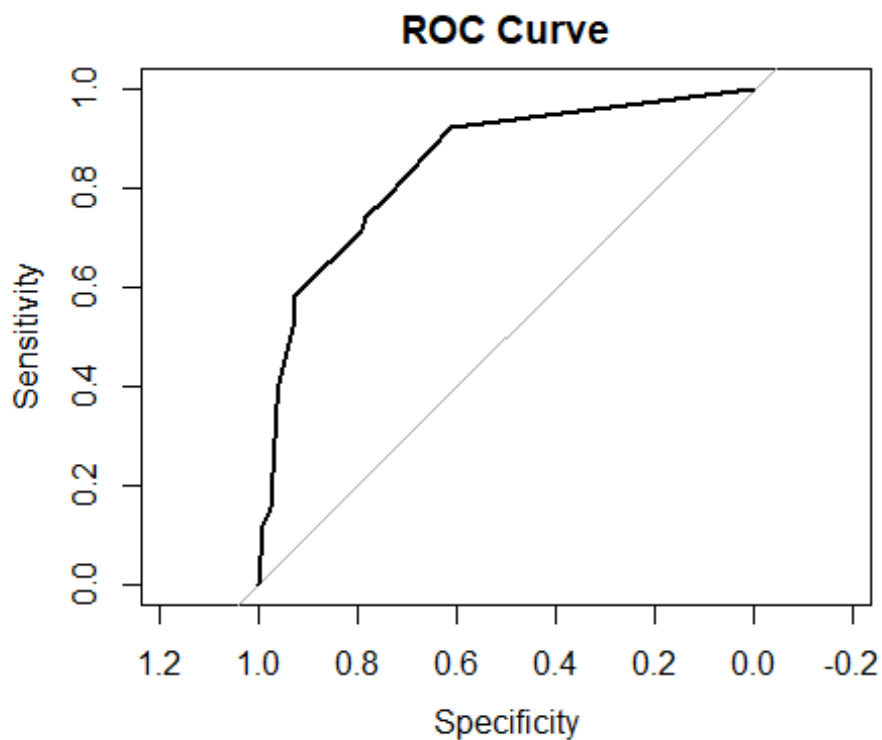
## Accuracy: 0.826087

```

```

cat("Precision: ", precision, "\n")
## Precision: 0.8483607
cat("Recall: ", recall, "\n")
## Recall: 0.9324324
cat("F1 Score: ", f1_score, "\n")
## F1 Score: 0.888412
#ROC Curve
pred_probs <- predict(dt_model, test_data, type = "prob")[,2]
roc_curve <- roc(test_data$Churn, pred_probs)
## Setting levels: control = No, case = Yes
## Setting direction: controls < cases
plot(roc_curve, main = "ROC Curve")

```



```

cat("AUC: ", auc(roc_curve), "\n")
## AUC: 0.8475196
#Hyperparameter tuning
tune_grid <- expand.grid(cp = seq(0.01, 0.1, by = 0.01))

```

```

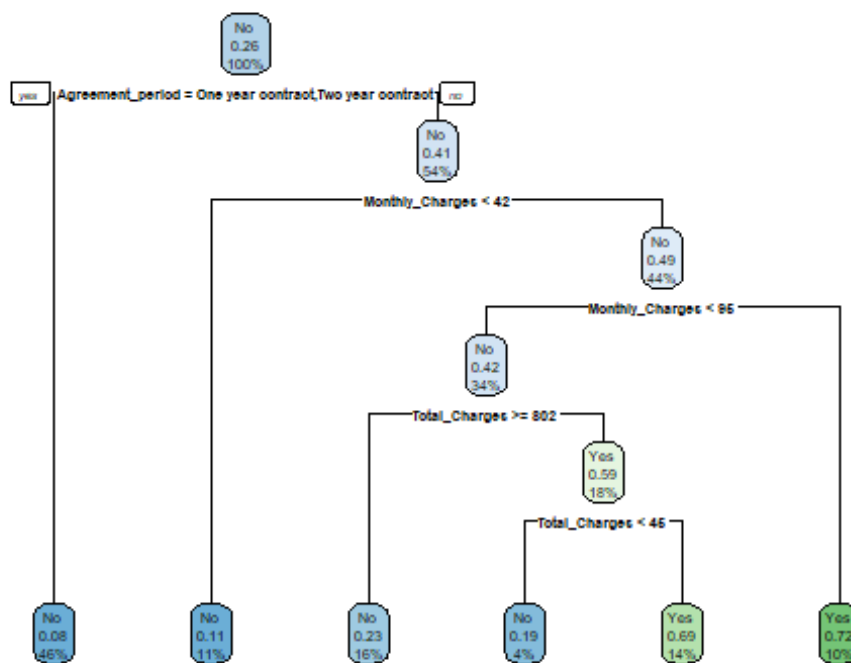
tune_model <- train(Churn ~ ., data = train_data, method = "rpart",
  trControl = trainControl(method = "cv", number = 10),
  tuneGrid = tune_grid)

print(tune_model)

## CART
##
## 701 samples
## 13 predictor
## 2 classes: 'No', 'Yes'
##
## No pre-processing
## Resampling: Cross-Validated (10 fold)
## Summary of sample sizes: 630, 631, 632, 631, 631, 630, ...
## Resampling results across tuning parameters:
##
##   cp    Accuracy   Kappa
##   0.01 0.7873330 0.41136908
##   0.02 0.7888225 0.43673828
##   0.03 0.7874957 0.41392658
##   0.04 0.7903528 0.42344564
##   0.05 0.7489243 0.15809132
##   0.06 0.7417814 0.10904616
##   0.07 0.7375561 0.07065137
##   0.08 0.7403919 0.00000000
##   0.09 0.7403919 0.00000000
##   0.10 0.7403919 0.00000000
##
## Accuracy was used to select the optimal model using the largest value.
## The final value used for the model was cp = 0.04.

#Pruning decision tree
pruned_model <- prune(dt_model, cp = 0.02)
rpart.plot(pruned_model)

```

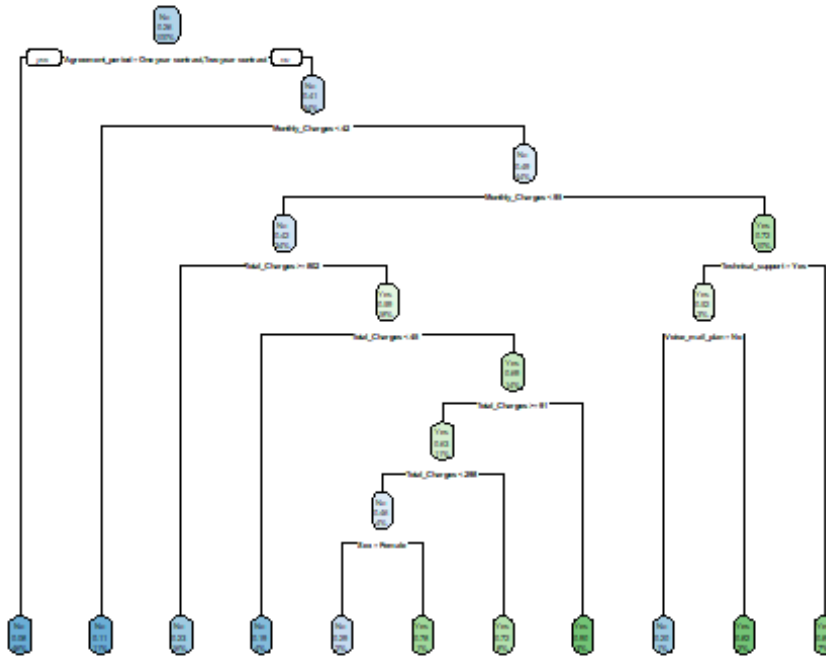


```
importance_caret <- varImp(tune_model, scale = FALSE)
print(importance_caret)
```

```
## rpart variable importance
##
##   only 20 most important variables shown (out of 25)
##
##                                     Overall
## Monthly_Charges                     68.870
## Internet_serviceNo Internet        42.477
## Technical_supportNo internet       42.477
## Term                               37.957
## Total_Charges                       32.250
## Streaming_VideosNo internet        27.102
## Agreement_periodOne year contract  21.132
## Agreement_periodTwo year contract  18.918
## Marital_StatusSingle               10.600
## Technical_supportYes                 4.189
## Multiple_lineYes                     2.095
## Voice_mail_planYes                   0.000
## International_planyes                0.000
## `Agreement_periodTwo year contract` 0.000
## Internet_serviceDSL                 0.000
## International_planYes                0.000
## Streaming_VideosYes                  0.000
## SexMale                             0.000
```

```
## `Technical_supportNo internet` 0.000
## `Internet_serviceFiber optic` 0.000
```

```
rpart.plot(dt_model)
```



This section focuses on developing a decision tree to predict customer churn. The process begins by splitting the dataset into training and testing sets to ensure a balanced evaluation of model performance. The model is then trained to classify customers as likely to churn or not based on attributes such as contract type, technical support, monthly charges, and more.

Once trained, the model is evaluated using several performance metrics. It achieved an accuracy of 82.6%, with a recall of 93.2%, indicating strong capability in identifying customers who are likely to churn. The model's precision and F1 score balance accuracy with the ability to minimize false positives and negatives. Hyperparameter tuning is conducted to optimize the complexity parameter (cp), selecting the best tree structure to avoid overfitting. Pruning further refines the tree, making the model less complex and more interpretable without sacrificing predictive power.

An ROC curve analysis provides an AUC score of 0.85, reinforcing the model's effectiveness in distinguishing between churn and non-churn cases. Important variables identified include total charges, contract length, technical support availability, and monthly charges, highlighting the key factors influencing customer decisions.

In conclusion, the decision tree model successfully identifies key drivers of customer churn, delivering solid predictive performance with an accuracy of 82.6% and an AUC score of 0.85. Insights gained from the model suggest that customers with shorter contracts, higher total charges, or limited technical support are more likely to churn. These results point to actionable strategies for customer retention, such as offering extended contract options, optimizing pricing, and enhancing technical support services. The model's interpretability allows for targeted churn-reduction initiatives, making it a valuable tool for strategic planning in customer retention.